Muhammad Hamza

Aspiring Software Quality Assurance Engineer

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LinkedIn | GitHub | Portfolio

Computer Science graduate with hands-on experience in Software Quality Assurance through academic projects and practical training. Familiar with manual testing, test case design, bug reporting, API testing using Postman, and end-to-end test automation using Playwright. Experienced in using JIRA (AIOTests) for defect tracking and working with enterprise platforms such as Microsoft Dynamics 365 and Microsoft Power Platform. Looking to start my career in Software Quality Assurance, where I can continue learning and contribute to building reliable software.

EDUCATION

FAST NUCES 2022 — 2026

Bachelor of Science in Computer Science

SKILLS

SQA

- Manual Testing
- Functional Testing
- Regression Testing
- Smoke Testing
- API Testing
- Test Case Design
- Bug Lifecycle Management
- SDLC & STLC

Tools & Platforms

- JIRA (AIOTests)
- Playwright
- Postman
- MS Dynamics 365
- MS Power Platform
- Git / GitHub
- VS Code

Programming Languages

- C / C++ (OOP)
- Python
- JavaScript
- Assembly (MASM)

Web Development

- HTML / CSS
- JavaScript
- React.js
- Python (Flask)

Backend & Databases

- RESTful APIs
- MySQL
- CRUD Development

PROFESSIONAL EXPERIENCE

PPC Landing Page Developer — Freelance Mid 2025 – Early 2026

- Delivered 100+ client landing pages with systematic cross-browser and responsive testing before every handoff.

SQA PROJECTS

Playwright E2E Test Automation — Blood Donation System Jun 2026

Playwright · JavaScript · Allure Reporting

- Built a Playwright-based SQA test suite for a Flask blood donation application, covering 40 end-to-end scenarios across positive and negative flows.
- Test coverage included login, registration, search, donor, hospital, and admin workflows with structured execution reports.
- Implemented Allure reporting and screenshot capture for clear, reproducible test documentation.

ACADEMIC PROJECTS

CleanAI — AI-Powered Waste Classification (FYP) Aug 2025 — May 2026

React.js · Next.js · Python · YOLO · Flask

- Developed and deployed an AI-powered waste sorting MVP using YOLO-based object detection for automated waste classification.
- Designed and deployed a responsive landing page, backend services, and admin dashboard using Vercel and Railway.
- Conducted pilot testing and performed unit, integration, smoke, and regression testing to validate functionality across the full system.
- Continuously trained and optimized the detection model for improved real-world accuracy.

Online Blood Donation Management System
Python (Flask) · MySQL

- Developed a Flask-based Blood Donation Management System with role-based authentication for donors, hospitals, and administrators.
- Implemented core features including profile management, blood request handling, appointment scheduling, donor search, notifications, and admin moderation.
- Performed deployment and Playwright-based test automation to improve application quality and reliability.

Atomic Chess — React, JavaScript (AI-Based Game Logic)

- Developed an interactive web-based chess variant implementing custom game rules, move validation, and explosion mechanics.
- Built the UI in React with clean state management and smooth animations. Integrated basic AI logic and tested gameplay for rule accuracy and stability.

Other Academic Projects

- Library Management System (C++ OOP) — Modular system with book inventory, member records, and borrowing workflows.
- Poker Game (C) — Terminal-based game with card shuffling, betting logic, and data structures.
- Regex-Based Search Engine (Python) — Lightweight text search tool using regular expressions for pattern matching.
- Hangman Game (Assembly 8086) — Fully interactive game using low-level instructions with string manipulation and game-state logic.